GROUP WEEKLY REPORTS

Project Title: Why 2024 Flights Ran Late
Team: Team 10 – DATA-VIZ HACK-A-BIT
Team Members: Annalakshmi R, Ashvitha I, Dharshini P, Joshna Amirthaa Mariappan

---

WEEK 2 – DATA COLLECTION AND DATASET UNDERSTANDING

During the second week, our team concentrated on collecting the required data and understanding its structure and characteristics. We obtained the 2024 flight dataset from Kaggle, which contains a large volume of information related to airline operations throughout the year.

The dataset consisted of approximately 7.08 million flight records and 35 columns, making it suitable for large-scale data analysis. We examined the dataset to understand the different attributes and their relevance to our project.

We explored important variables such as airline carrier, origin airport, destination airport, scheduled departure time, actual

departure time, departure delay, scheduled arrival time, actual arrival time, arrival delay, flight duration, and distance.

Our team discussed how these variables could be used to investigate flight delays and identify meaningful patterns. We also examined the distinction between scheduled and actual flight timings and understood how departure and arrival delays are calculated.

We used Python and the Pandas library to load and inspect the dataset. Functions such as `read_csv()`, `head()`, `shape`, `info()`, and `describe()` helped us understand the dataset's size, structure, data types, and statistical properties.

We also identified challenges associated with handling a dataset of this size, including memory usage, processing time, and the need for efficient data manipulation techniques.

By the end of the week, we had developed a clear understanding of the dataset and identified the variables required for further analysis. This stage provided the foundation for the subsequent data quality assessment and preprocessing activities.